

WASSIM FARHAT

SKILLS

- Electrical Troubleshooting
- Problem Solving
- Critical Thinking & Root Cause Analysis
- Electrical Safety & Standards
- Preventive & Corrective Maintenance
- Technical Documentation & Communication

EXPERTISE

TIA Portal · WinCC · PID Control · Arduino · STM32 · MQTT · AutoCAD · C++ · Altium Designer · MATLAB · Simulink · VHDL

LANGUAGES

Arabic	Native
English	Advanced
French	Intermediate
German	Beginner

CERTIFICATIONS

-English B2

-IT Specialist AI

-MATLAB, SIMULINK, SIMSCAPE

-Ethics & Innovation

Electrical & Automation Engineering Student

+21692983401 wassim.farhat@polytechnicien.tn
Hammam Sousse, Sousse

SUMMARY

Electrical Engineering student with hands-on experience in industrial automation, electrical systems, and electromechanical equipment. Skilled in troubleshooting, preventive maintenance, PLC/SCADA systems, electrical installations, and control systems. Proficient in TIA Portal, WinCC, Embedded Systems, MATLAB/Simulink, and AutoCAD, with a strong problem-solving mindset and focus on safety, reliability, and continuous improvement.

EXPERIENCE

Process Engineering Intern
08/2026 - 09/2026

DRÄXLMAIER — Sousse

- Analyzed MES production data across 3 automated crimping/cutting machines to trace scrap contribution by machine 25–37.5% split.
- Diagnosed root causes using QQOQCP, Pareto analysis, Ishikawa diagrams, and AMDEC pinpointing blade wear, calibration, and feed-parameter issues.
- Proposed and drove corrective actions (preventive maintenance intervals, pre-startup checklist, operator training) that delivered a measurable scrap-rate reduction 7-10%.

Electrical & Solar PV Installation Intern
07/2026 - 08/2026

Ouerdani Electricity Corporation — Sousse

- Designed AutoCAD schematics for residential electrical installations and wired panels according to NF C 15-100 standards,including grounding & breaker sizing.
- Sized AC circuit breakers using load calculations and wired lighting, socket, and DCL circuits on-site.
- Assisted on a full grid-connected solar PV install structural mounting, DC/AC wiring, and commissioning under Tunisia's ANME/STEG framework.

IoT Technician Intern
06/2025 - 07/2025

Sitex — Sousse

- Replaced a manual generator control system with an ESP32-based controller, integrating 6 sensors for temperature, oil pressure, fuel level, battery voltage, AC current, and 3-phase voltage.
- Streamed data from 6 sensors via firebase and developed a MIT app for real-time remote monitoring and control.

Electrical Automation Technician Intern(PFE)
01/2024 - 05/2024

Sitex — Sousse

- Migrated a legacy wired system to Siemens S7-1500 (TIA Portal V17) after full I/O analysis across 7 machines, wiring relay circuits and sizing Digital & Analog I/O modules.
- Designed 14 GRAFCET diagrams (auto, manual, security mode) and implemented structured PLC blocks (OB/FB/FC/DB) with analog scaling and PID control.
- Built a WinCC HMI for real-time machine-state monitoring and alarm supervision, reducing maintenance delays by 12%, plus a full electrical wiring diagrams for the control cabinet.

PROJECTS

PCB Design, Altium Designer
05/2025

- Built an integrated component library from datasheets: schematic symbols, footprints and 3D STEP models, including a 14-pin quad op-amp (AD8630).
- Converted a complex paper schematic into a multi-sheet project, generated the PCB, and hand-routed it following length, spacing and track-width rules.

Exported the PCB layers ready for fabrication.

Smart Parking with Biometric Access 05/2025

- Developed an Arduino Uno access-control system with two-factor entry (fingerprint, then PIN keypad) driving servo-controlled barriers.
- Added PIR-based automatic exit detection and an I²C LCD showing available places in real time.
- Programmed the control logic and integrated the sensors, the servos and the display.

Mind Makers: Arduino Workshop for Teenagers 04/2025

- Organised and led hands-on electronics and robotics workshops for 22 students aged 12 to 15, in partnership with Future Vision.
- Designed activities around Arduino experiments and a final car race. 82% of participants reported being highly satisfied.

EDUCATION

Electrical and Automation Engineering 2024 - Present

Polytechnique Sousse Sousse

Bachelor's in Electrical and Automation 2021 - 2024

Higher School of Science and Technology Hammam sousse Sousse